

UNIT-5

MACHINE LEARNING

- Support Vector Machines
- Bayesian learning
- Application of machine learning in
 - computer vision
 - speech processing
 - Natural language processing.

Support Vector Machines

- Support Vector Machine (SVM) is a powerful supervised machine learning algorithm used mainly for classification problems and also for regression tasks.

The main goal of SVM is:

- To correctly classify data points, and
- to ~~maximize~~ the maximize the distance between the classes and the separating boundary.

Explanation → Suppose we have two categories of data.

- Class A
- Class B

SVM draws a line in 2D, plane in 3D or hyperplane (in higher dimensions) to separate these classes.

But SVM does not choose any random boundary. It chooses the boundary that gives the maximum margin.

Support Vectors.

The data points nearest to the hyperplane are called support vectors.

These ~~two~~ points are very important because:-

- They determine the position of the hyperplane.
- Removing them changes the decision boundary.

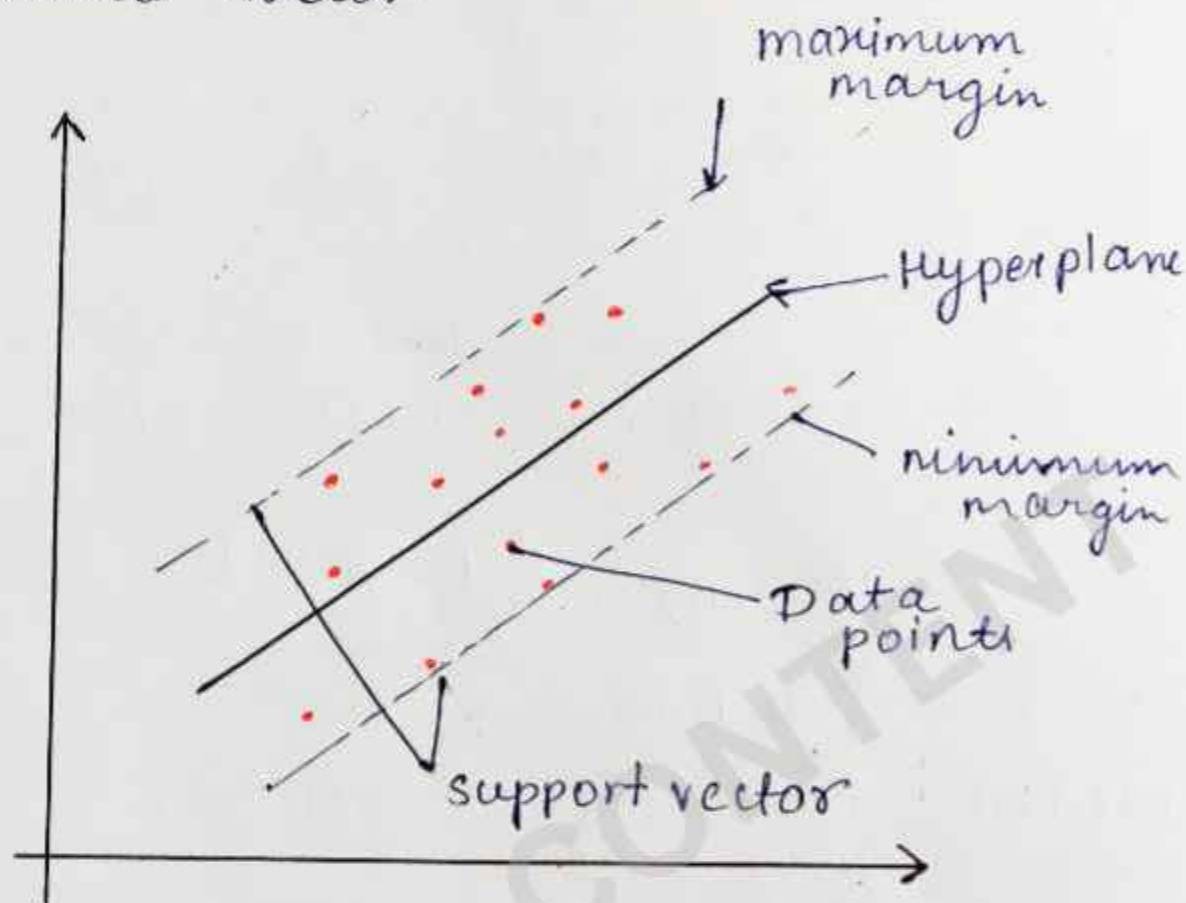
• SVM mainly depends on these support vectors instead of the entire dataset.

Types of SVM

① Linear SVM → When data can be separated using a straight line, SVM is called linear SVM.

② Non-Linear SVM → Sometimes data cannot be separated using straight line. In such cases, SVM uses the kernel Trick.

The kernel converts low-dimensional data into higher-dimensional space where separation becomes easier.



Support Vector Machines.

Application of SVM

- Face detection
- Handwriting recognition
- Medical diagnosis
- Fraud detection.
- Image classification

BAYESIAN LEARNING

Bayes' theorem :- Bayes' theorem is a mathematical formula used to find the probability of an event based on prior knowledge and new evidence.

It is widely used in ML, medical diagnosis, AI, Data science, Spam filtering.

Formula $\rightarrow$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Meaning of Terms

- $P(A|B) \rightarrow$ Probability of A happening when B has already happened.

It is called Posterior Probability.

- $P(B|A) \rightarrow$ Probability of B happening if A is true.

It is called Likelihood.

- $P(A) \rightarrow$ Probability of A before seeing evidence.
It is called prior probability.

- $P(B) \rightarrow$ Total Probability of B

It is called evidence.

Bayesian Learning:-

Bayesian learning is a machine learning method based on Bayes' theorem.

It learns by

- starting with an initial belief
- Observing data
- updating the belief using probability.

In simple words:-

Bayesian learning means "learning from experience by updating knowledge again and again."

The mathematical rule is:-

$$P(H|D) = \frac{P(D|H) P(H)}{P(D)}$$

where:-

- $P(H)$: Prior probability $\rightarrow$ belief before seeing data.
- $P(D|H)$: Likelihood - probability of observing the data if the hypothesis is true.
- $P(D)$: Evidence:- overall probability of data.
- $P(H|D)$: Posterior probability $\rightarrow$ update belief after seeing data.

In ML, Bayesian learning treats model parameters as probability distributions rather than fixed values.

Instead of

"the parameter is exactly 2.5"

Bayesian learning says:

"the parameter is probably around 2.5 with some uncertainty."

Concept of posterior Probability.

In Bayesian learning, posterior probability means:

- updated probability after seeing new data or evidence.
- It is the final probability we get after applying Bayes' theorem.

Formula
$$P(H|D) = \frac{P(D|H) P(H)}{P(D)}$$

Here:-

$P(H|D)$ → Posterior Probability

H → Hypothesis

D → Data / Evidence.

So, Posterior probability = probability of hypothesis after observing data.

Advantages of Bayesian Learning

1. Handles uncertainty. Provides probability instead of only yes/no answers.
2. Learns from small data. Works well even when data is limited.
3. Continuously Improves. Updates knowledge whenever new data arrives.
4. Uses Prior knowledge $\therefore$ can use already known information.

Disadvantages of Bayesian Learning.

1. Computationally Expensive. Calculations becomes difficult for very large data.
2. Choosing Prior is Difficult. Wrong prior knowledge can affect result.
3. Complex Mathematics. Advanced Bayesian models requires strong probability knowledge.

Application of Machine Learning in COMPUTER VISION

Machine Learning in Computer Vision

→ Machine Learning helps computer vision systems learn pattern from images and improve automatically using data.

Computer Vision → It is a field of AI that enables computers to understand and analyze images and videos like humans.

Application /

Role of Computer Vision in ML

① Image Understanding → Computer vision helps ML models understand image content.

eg → identifying cats, dogs, cars or humans in photos.

It learns patterns from many images and improves accuracy over time.

② Object Detection → It helps to detect and locate objects inside images or videos.

eg → detecting vehicles on roads.

- detecting faces in photos.

- detecting tumors in scans.

3 feature ~~Detectio~~ Extraction

computer vision extracts information important features from images such as:-

- edges
- shapes
- colors and textures.

And ML use these features for prediction & classification.

4. Medical Image Analysis

Computer vision helps ML analyze.

- X-rays
- MRI scans
- CT scans

⑤ Pattern Recognition

ML models learn visual patterns using computer vision.
like handwriting recognition.

- Fingerprint recognition
- Facial Recognition

⑥ Automation of Visual Tasks

Tasks that normally requires human eyes can be automated.

Example - Quality Inspection in factories

- Traffic monitoring
- Surveillance systems.

This reduces human effort and increases speed.

⑦ Technologies Used

Common ML techniques used in computer vision.

- Neural Networks
- Convolutional Neural Network (CNN)
- Deep Learning
- Image Processing
- Pattern Recognition.

Application of ML in Speech Recognition.

Speech Recognition → Speech Recognition is the technology that converts spoken language into text or commands using algorithms and machine learning.

It is used in things like:

- Voice assistants such as Siri, Google Assistant and Alexa
- Dictation software
- Call center automation.
- Real-time transcription

Working steps →

1. A microphone captures speech
2. then background noise is filtered and speech is digitized.
3. The system identifies speech patterns
4. AI predicts likely words and sentences.
5. The recognized speech becomes text or triggers commands.

Example → if you says:

"set an alarm for 7 AM"

* speech recognition converts it to text, and another system interprets the intent and sets the alarm.

Application of ML in speech Recognition.

ML plays a major role in speech recognition systems. It helps computers understand, process and convert human speech into text accurately.

Important Application given below.

1] Virtual Voice Assistants → ML helps assistant understand voice commands and respond intelligently.

e.g → Google Assistant, Siri, Alexa.

Uses →

- Setting Alarms

- Playing music

- Searching the internet

- Sending messages.

2] Speech - to - Text Conversion

ML converts spoken languages into written text.

- Automatic subtitles
- Meeting transcription
- Online classes notes
- Dictation software

Eg, Doctors and journalists use speech-to-text tools to save time

3] Customer Service Automation

Many companies use ML-based speech recognition in call centers.

Uses in

- Automated customer support
- IVR system ["press 1 for English"]
- Voice based complaint systems

Benefits :-

- Faster Response
- Reduced human workload
- 24/7 service

4] Language Translation

ML systems recognize speech in one language and translate it into another language.

Example: English speech → Hindi text / audio.

Used in:

- Travel apps
- International meetings
- Online communication

5] Accessibility for Disabled People.

Speech recognition helps people with physical disabilities interact with devices using voice.

used in

- Voice typing
- Hands-free computer control
- Reading and writing assistance

6] Smart Home Control

ML allows users to control smart devices using voice commands.

Eg

- Turn on light
- Control TV
- Adjust AC Temperature.

Application of ML in Natural Language Processing.

Natural Language Processing (NLP)

It is a branch of AI and ML that enables computers to understand, interpret, process, and generate human language.

Human Language includes: Text, Speech, Sentences, Emotions, Grammar

NLP helps computers communicate with human in natural ways.

NLP is the technology that allows computers to understand and work with human language such as English, Hindi etc.

Components of NLP.

①. Natural Language Understanding (NLU)
Helps the computer understand the meaning of language.

Example "Book a ticket to Delhi"

The system understands.

- action = book
- destination = Delhi.

② Natural Language Generation.

Helps the computers generate human-like responses

Examples - chatbots replying to users.

Application of ML in NLP.

1. Machine Translation

ML translates text from one language to another automatically.

Eg →

- English → Hindi
- Hindi → French.

Applications →

- Travel apps

- International communication
- Online learning

2. Chatbots and Virtual Assistants.

ML enables chatbots to understand user questions and respond intelligently.

Example → chatgpt.com

- Google Assistant
- Siri

Used in

- Customer care
- Online shopping
- Banking Assistance

3] Sentiment Analysis

ML detects emotions or opinions from text.

Example → This movie is excellent = Positive sentiment.

Used in:-

- Product review analysis
- Social media monitoring
- Brand Reputation tracking

4] Speech Recognition.

ML converts spoken ~~text~~ language into text.

Used in

- Voice typing
- Voice Assistants
- Automatic subtitles.

Example

→ "Call Mom" → converted into text/command.

5] Text Summarization.

ML creates short summaries from long documents.

Used in

- News summarization
- Research papers
- Meeting notes